



# FOREWORD

When my brother and I were growing up, our mom taught us how scientists answer their questions. They use a process called the scientific method. After this, we spent a lot of time in the kitchen doing classic experiments, just like the ones you'll find in this book.

Before we embarked on an experiment, we always started with a question: How can fruit become a battery? What do I need to include in a water filter? How can I bake ice cream and not have it melt into a gooey mess? Then we read up on our topic, we did our background research, and came up with an educated guess of what was going to happen in our experiment. This task is called constructing a hypothesis, or making an educated guess, and it's essential to the scientific process.

Then it was time to test that hypothesis by conducting the experiment and seeing what happened. Scientists

call this making observations. Did the end results of the experiment agree with the hypothesis? What if one thing in the experiment were changed? Each thing that can be changed in an experiment is called a variable. For instance, what would happen if you used a different fruit when making your battery? What if you modified the design of your airplane wing or increased the amount of shampoo you put in your slime?

My brother and I enjoyed working together on experiments like the ones featured in *Maker Lab*—coming up with our own hypotheses, trying out our ideas, and comparing results. And, of course, we always liked sharing what we learned. You, too, can share your scientific research with a family member, friend, or teacher. You are a scientist!

Every scientist needs to know the basics of the scientific method, which consists of the following steps: ask

